Experiment No. 4	
SE (CSE)	ROLL NO :
Date of Implementation:	
Aim: Implement simple linear regression for the given data set	
Programming Language Used : Python	
Upon completion of this experiment, students will be able to CO3 : Perform prediction by applying regression models	

Indicator	Poor	Average	Good
Timeline Maintains submission deadline (2)	Submission not done (0)	One or More than One week late (1)	Maintains deadline (2)
Completion and Organization (2)	Many mistakes in documents (0)	Document is just acceptable (1)	Completed whole document and neatly organized (2)
Program Performance (2)	Could not perform at all (0)	Implemented few parts (1)	Full implementation (2)
Knowledge In depth knowledge of the Experiment (4)	Unable to answer questions(0)	Unable to answer few questions (1-2)	Able to answer all questions with proper explanation (3-4)

Assessment Marks :

Timeline	
Completion and Organization	
Program Performance	
Knowledge	

TOTAL :

EXPERIMENT	***4***
Aim	Implement simple linear regression for the given data set
Theory	Simple linear regression is used to estimate the relationship between two quantitative variables. You can use simple linear regression when you want to know: 1. How strong the relationship is between two variables (e.g., the relationship between rainfall and soil erosion). 2. The value of the dependent variable at a certain value of the independent variable (e.g., the amount of soil erosion at a certain level of rainfall). Regression models describe the relationship between variables by fitting a line to the observed data. Linear regression models use a straight line, while logistic and nonlinear regression models use a curved line. Regression allows you to estimate how a dependent variable change as the independent variable(s) change. Simple linear regression is a parametric test, meaning that it makes certain assumptions about the data. These assumptions are: Homogeneity of variance, Independence of observations, data follows a normal distribution and relationship between the independent and dependent variable is linear.
Implementation	1. Perform the given task in python Consider the following data (see table below) a) Print first 5 rows of data b) Display scatter plot of data c) Calculate (using formula) and print regression coefficients b0 and b1 d) Display regression line equation e) Calculate and print coefficient of determination (R squared, Residual sum of squares (RSS), and RMSE f) Plot regression line g) Predict the value of y given x=10 2. Perform the following task using inbuilt functions in python a) Import the packages numpy, pandas and the sklearn.linear_model b) Read data set (advertising.csv) c) Select the column 'TV' as independent variable and 'sales' as dependent variable d) Divide data into training and testing split e) Create an instance of the class LinearRegression, which will represent the regression model f) Fit the model for training data g) Get coefficients of regression and coefficient of determination from the model h) Apply the model for predictions on testing data.

X	0	1	2	3	4	5	6	7	8	9
y	1	3	2	5	7	8	8	9	10	12

	i) Show the residual error plot for training (with green color dots), testing data (blue color) and zero residual error line
Conclusion	Comment on the predicted output and the corresponding error value obtained. Additionally, compare the regression coefficients calculated manually with those computed using built-in functions
Post Lab Questions:	1. Explain standard error of coefficients 2. Explain coefficient of correlation, coefficient of determination